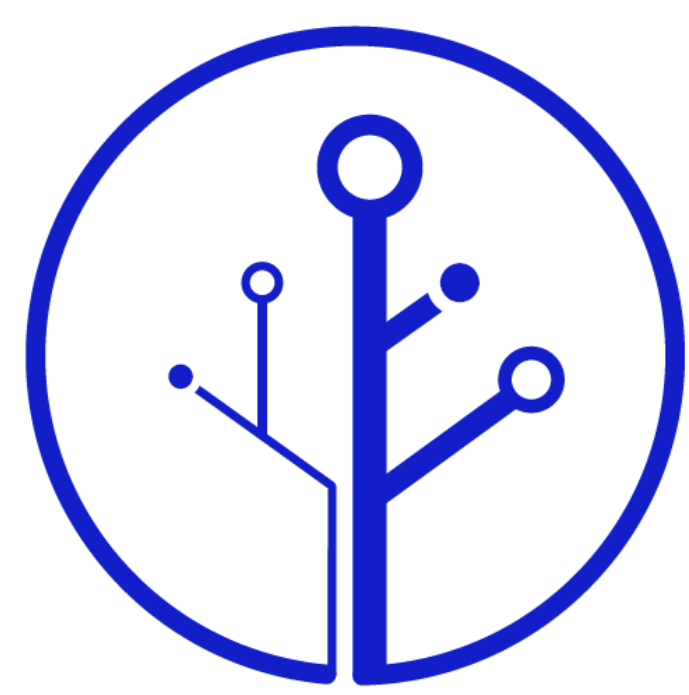




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# EVALUATING METHODS FOR BIOMARKER CUT-OFF DETECTION: A COMPARATIVE STUDY IN SMALL AND UNBALANCED SAMPLES

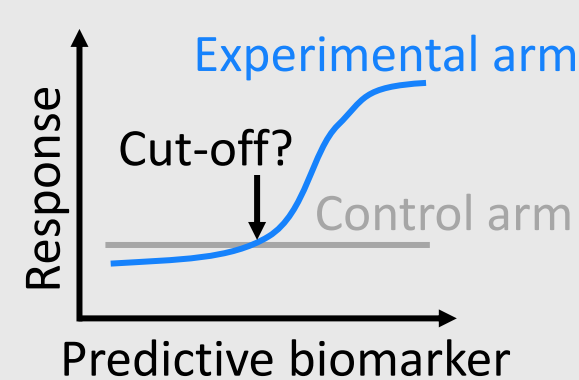
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## Introduction

### Definition: predictive biomarkers

- Biomarkers are measured biological characteristics and key metrics to evaluate treatments.
- Predictive biomarkers can **predict the response when they are associated with the treatment variable**.



### Definition: biomarker cut-offs

- The cut-off point of a predictive biomarker **divides patients into two subgroups**: patients below the cut-off are conventionally considered less likely to respond positively to treatment than the control, while patients with values above the cut-off are expected to exhibit higher response rates than the control.
- Choosing a reliable cut-off point is challenging in **small or unbalanced trials**.

### Objectives

- Main setting: 60 patients, 40:20 (experimental:control)**  
**1 biomarker (Uniform or Gamma), 1 treatment (2 arms), logistic probability of response**
- Compare **11 cut-off identification methods** (4 one-arm, 7 two-arm)
  - Assess bias, variance and practical behaviour** when samples are small and allocation is unbalanced
  - Give guidance on which methods are most robust** under different predictive scenarios and biomarker distributions (uniform and gamma)

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## Methods

### Methods considered

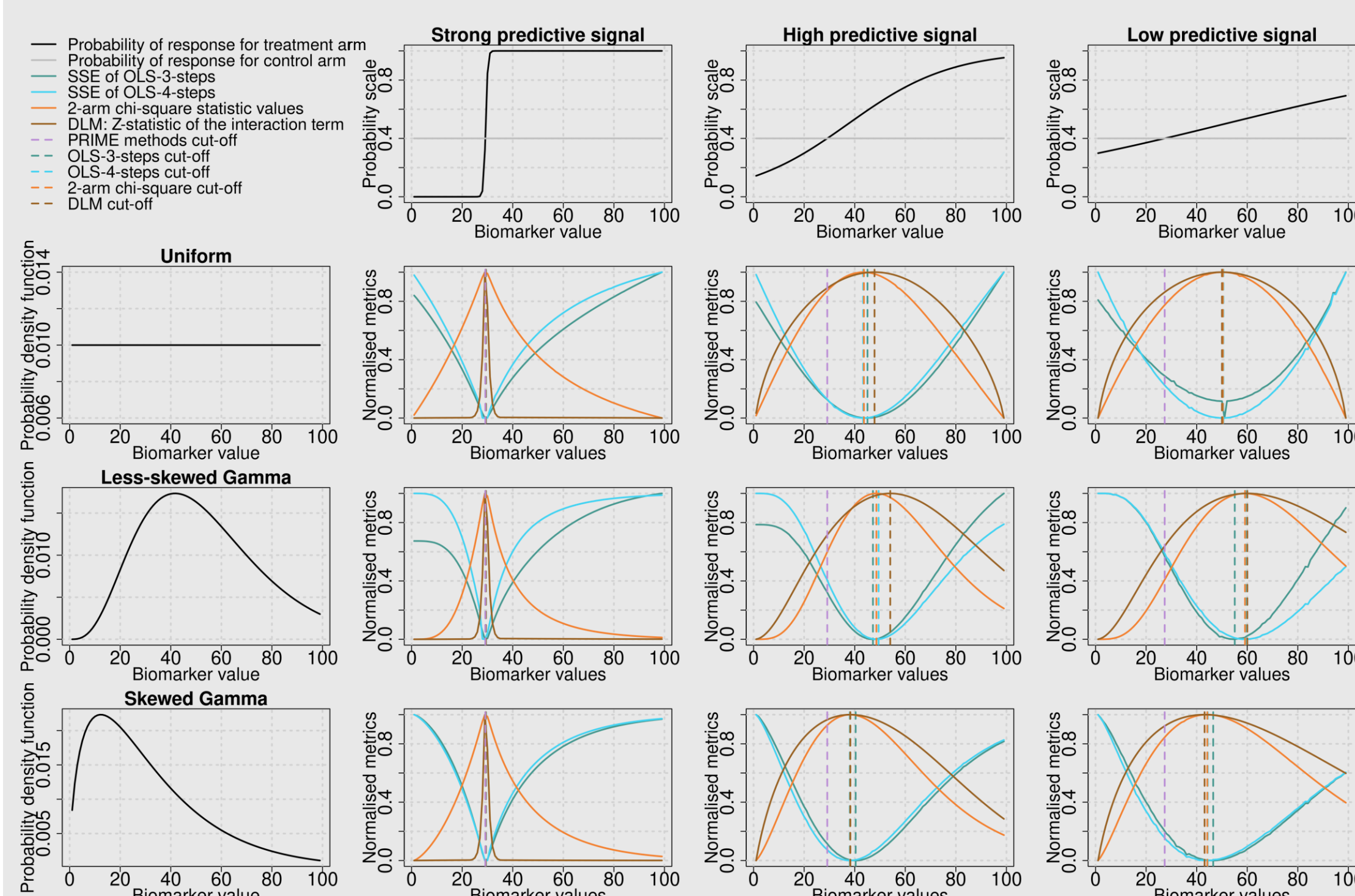
#### One-arm:

- Youden index ( $Se + Sp$ )**: Maximise the sum of sensitivity and specificity
- Product  $Se \times Sp$** : Maximise the product of sensitivity and specificity
- Step-function fitted with ordinary least squares (SF-OLS)**: Best step function fitting the data
- Maximum chi-squared statistic ( $\text{Max}(\chi^2)$ )**: Maximise chi-square statistic for biomarker subgroups vs response

#### Two-arm:

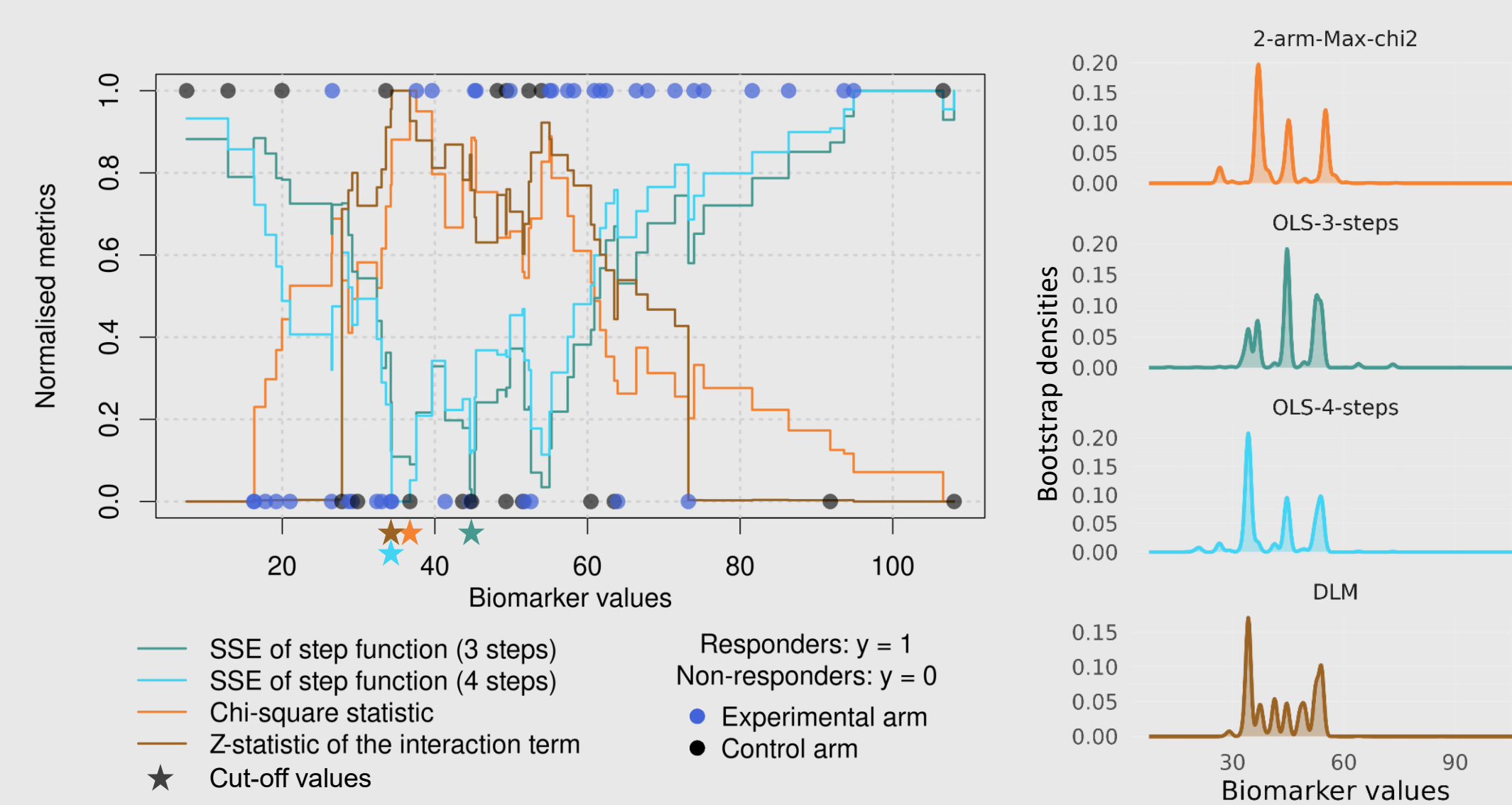
- Predictive biomarker graphical approach (PRIME)**: Model treatment effect as a function of biomarker and define cut-off where effect crosses zero
- Our proposal PRIME50% and PRIME80%**: PRIME variants where the cut-off is defined where the lower 50% (resp. 80%) confidence interval of the treatment effect crosses zero
- 2-arm  $\text{Max}(\chi^2)$** : Extension of  $\text{Max}(\chi^2)$  with treatment and subgroup vs response
- OLS-3-steps**: Extension of SF-OLS to three levels, with common level below threshold and split by arm above threshold
- OLS-4-steps**: Extension of SF-OLS to four levels, with split by arm below and above threshold
- Dichotomised logistic modelling (DLM)**: Best fit of logistic regression with dichotomised biomarker term

### Theoretical cut-off values derived from the definition of the two-arm methods



### Illustration on real data for 2-arm methods

60 patients, 40:20 (experimental:control)  
Biomarker values follow a **Gamma distribution**.  
 **$P(\text{response})$  is logistic** with treatment, prognostic (depends only on biomarker values and predictive (interaction between treatment and biomarker) terms).



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## Simulation study and results

### Setting

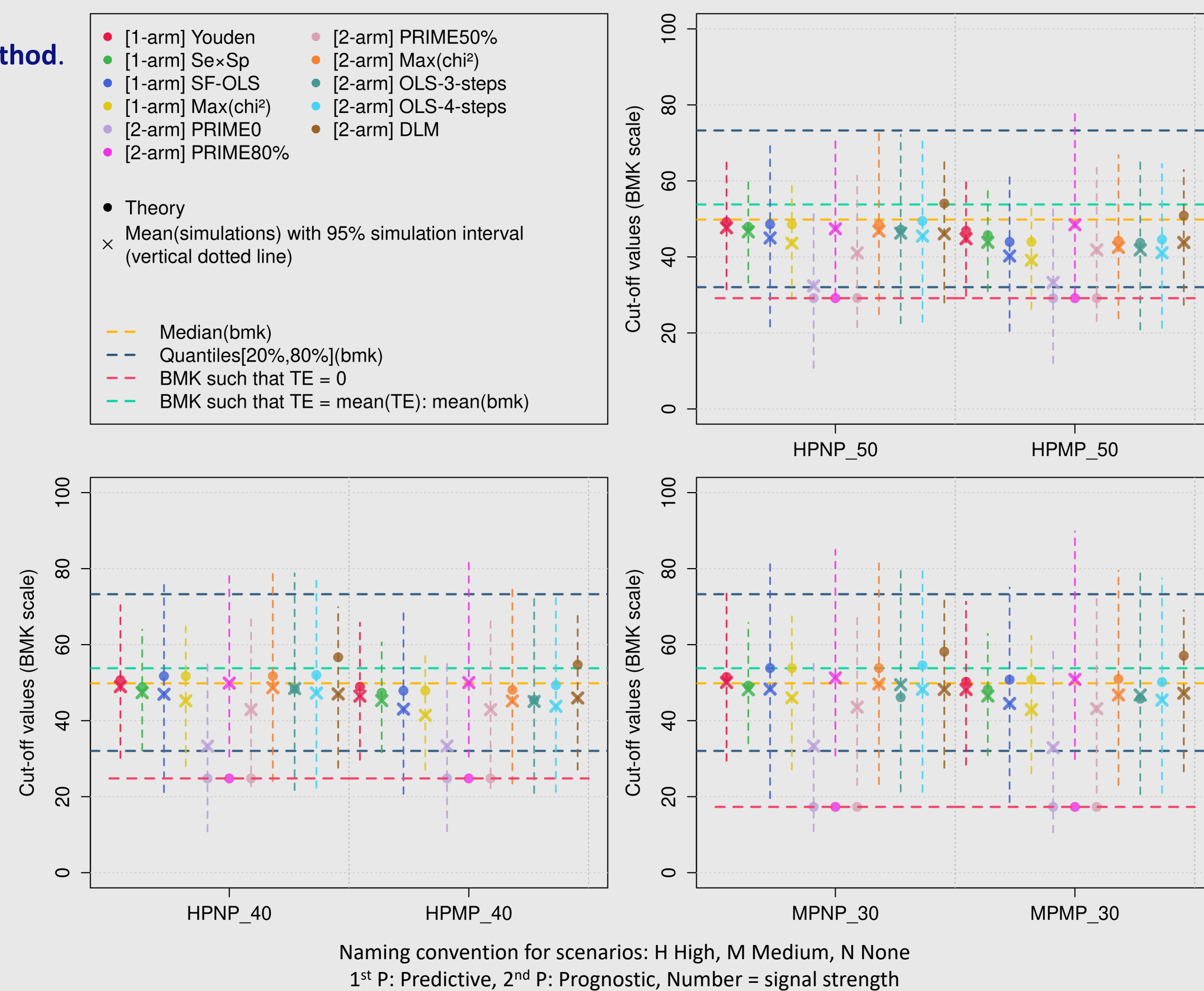
5,000 iterations per scenario. AKSA filter to assess **predictiveness** (see Serra et al. 2025) **before applying a cut-off identification method**.  
Main sample size at 40:20 and sample size sensitivity analysis: 40:40, 80:40, 80:80, 120:120  
6 scenarios with different strengths of predictive and prognostic signals

### Results (main setting)

| Metrics averaged across all scenarios | 1-arm methods |                |        |                      | 2-arm methods |          |          |                      |             |             |      |
|---------------------------------------|---------------|----------------|--------|----------------------|---------------|----------|----------|----------------------|-------------|-------------|------|
|                                       | Youden        | $Sp \times Se$ | SF-OLS | $\text{Max}(\chi^2)$ | PRIME0        | PRIME80% | PRIME50% | $\text{Max}(\chi^2)$ | OLS 3-steps | OLS 4-steps | DLM  |
| $\sqrt{\text{MSE}}$                   | 10.0          | 7.8            | 14.4   | 10.9                 | 15.3          | 29.2     | 22.1     | 14.2                 | 14.0        | 14.6        | 14.0 |
| Bias                                  | -1.9          | -1.6           | -4.8   | -6.4                 | 9.3           | 25.9     | 18.8     | -3.0                 | 0.2         | -4.8        | -8.9 |
| Standard deviation                    | 9.8           | 7.6            | 13.6   | 8.9                  | 11.6          | 13.3     | 11.3     | 13.8                 | 13.9        | 13.8        | 10.8 |
| $P(c > \text{bmk}_{\text{TE}=0})$     | 0.99          | 1.00           | 0.93   | 0.98                 | 0.76          | 0.99     | 0.95     | 0.96                 | 0.94        | 0.94        | 0.98 |
| Mean BMK+ proportion                  | 0.54          | 0.55           | 0.58   | 0.61                 | 0.76          | 0.52     | 0.63     | 0.56                 | 0.56        | 0.58        | 0.56 |
| $P(c > c_{\text{theory}})$            | 0.40          | 0.40           | 0.36   | 0.24                 | 0.76          | 0.99     | 0.95     | 0.39                 | 0.48        | 0.35        | 0.22 |

- For one-arm methods: **SexSp has the lowest  $\sqrt{\text{MSE}}$ , bias and SD across scenarios.**
- For two-arm methods: **OLS-3-steps has the lowest bias and DLM the lowest variance.**
- All one-arm methods underestimate the theoretical cut-off values.
- PRIME yields unique results**: it is the most interpretable method (the cut-off is defined where the estimated treatment effect crosses 0) but has high bias and large variance due to its model-based design: very sensitive in a small sample size setting. PRIME80% gives an 80% probability that the treatment effect is positive at the cut-off point.

### Average cut-off values found by methods and by scenarios with their theoretical values (main setting)



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## Key takeaways

- Recommendation**: assess predictiveness before applying a cut-off identification method.
- The determining factor for defining theoretical cut-off values is the strength of the predictive effect. In scenarios with a very sharp response rate, the choice of the method does not have an impact on the cut-off values found.
- When the response rate is mild, the differences between methods arise. The differences are more marked as the distribution of the biomarker values becomes skewed (considering one-arm and two-arm methods).
- The sample size sensitivity analysis shows that the use of a control arm introduces noise. All methods improve as the sample size increases, except non-PRIME two-arm methods which benefit more from a smaller sample size as their control arm.
- We propose PRIME50% and PRIME80% which are model-based methods that define a cut-off as the intersection between the lower confidence interval of the treatment effect estimated by the fitted model. These methods should not be compared to other methods only using  $\sqrt{\text{MSE}}$ , bias and SD, because of their high sensitivity to sample size. They provide interpretable results: picking a cut-off using PRIME80% reflects an 80% confidence that the treatment effect is positive at the cut-off point.